Mathematics 2 (5EZB0) Lecture 10

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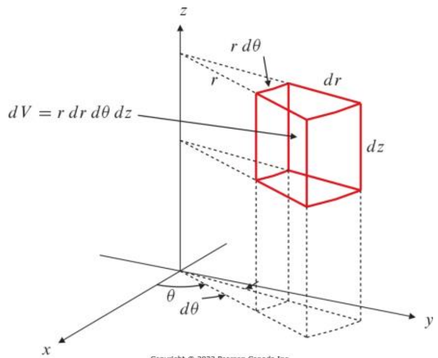
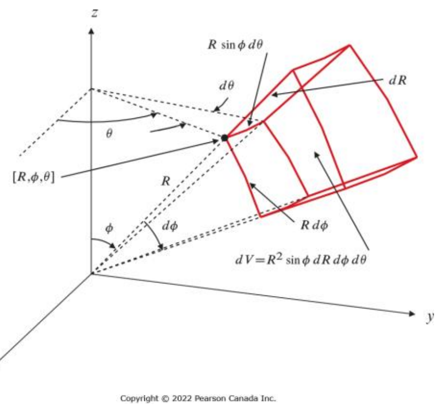
Change of variables





Now consider the change of variable

$$x = x(u,v,w)$$
$$y = y(u,v,w)$$
$$z = z(u,v,w)$$

- Let $f\big(x(u,v,w), y(u,v,w), z(u,v,w)\big) = h(u,v,w)$
- Let the domain $\mathcal{D}$ be transformed into S in (u,v,w) coordinates

$$\int\int\int_{\mathcal{D}} f(x,y,z)dxdydz = \int\int\int_{S} h(u,v,w)\left|\det\left(\frac{\partial(x,y,z)}{\partial(u,v,w)}\right)\right| dudvdw$$

Cylindrical Coordinates

- $x = r\cos\theta, y = r\sin\theta, z = z$
- $\left|\det\left(\frac{\partial(x,y,z)}{\partial(r,\theta,z)}\right)\right| = r$

Spherical Coordinates

- $x = R\sin\phi\cos\theta, y = R\sin\phi\sin\theta\,, z = R\cos\phi$
- $\left|\det\left(\frac{\partial(x,y,z)}{\partial(R,\phi,\theta)}\right)\right| = R^2\sin\phi$

Surface area

We saw earlier that a surface normal of $z = f(x, y)$ is along

$$\boldsymbol{n} = (\nabla f(x, y), -1)$$

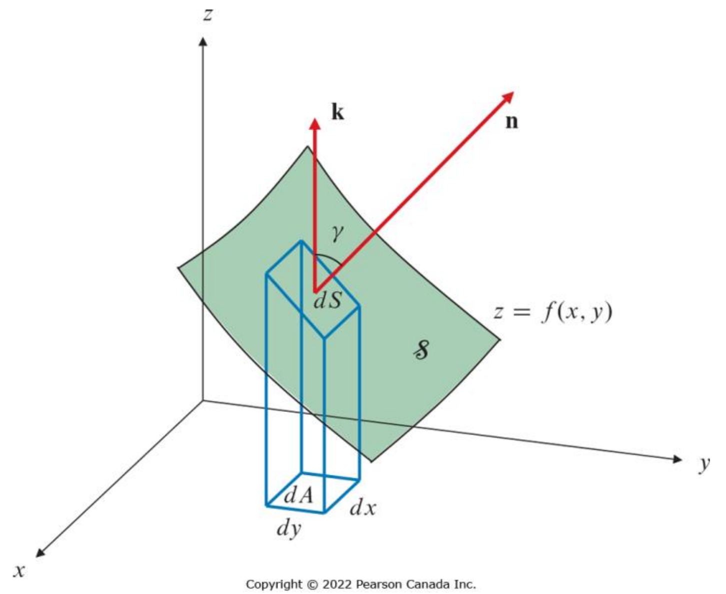
- The angle with the z-axis is then $\cos\gamma = \frac{|\boldsymbol{n}\cdot\boldsymbol{k}|}{|\boldsymbol{n}||\boldsymbol{k}|} = \frac{1}{\sqrt{1+|\nabla f|^2}}$

- The area of the surface element would be

$$d\mathcal{S} = \frac{dA}{\cos\gamma} = \sqrt{1 + |\nabla f|^2}\, dA$$

- Then the surface area is

$$\mathcal{S} = \int\int_{\mathcal{D}} \sqrt{1 + |\nabla f|^2}\, dA$$

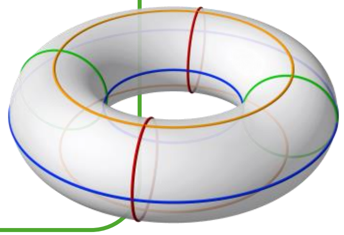


Surface of revolution

The surface area of the function
$z = f(r)$ $(r > 0)$ is $2\pi \int_0^\infty r\sqrt{1 + f'(r)^2}\, dr$

Question 1

Find the surface area of a Torus with major radius R, and minor radius b.



Revisiting Scalar & Vector fields

Vector fields

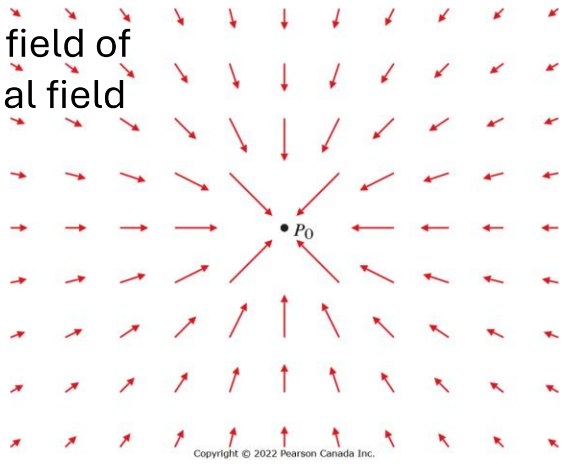
For a domain $\mathcal{D} \subseteq \mathbb{R}^d$, a vector field $\boldsymbol{f}$ is a function of the form

$$\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \dots, f_n(\boldsymbol{x}))$$

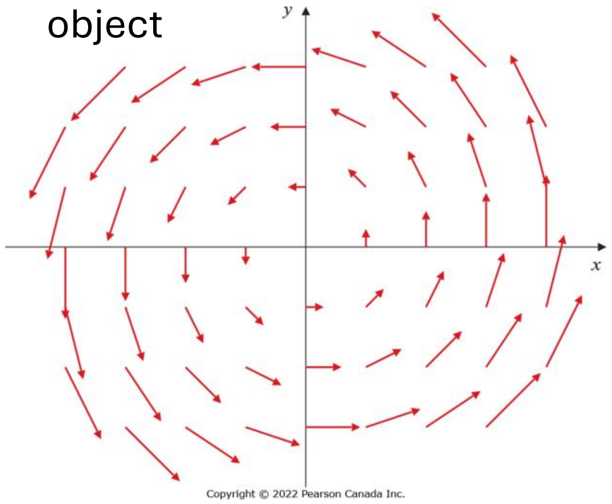
written also as

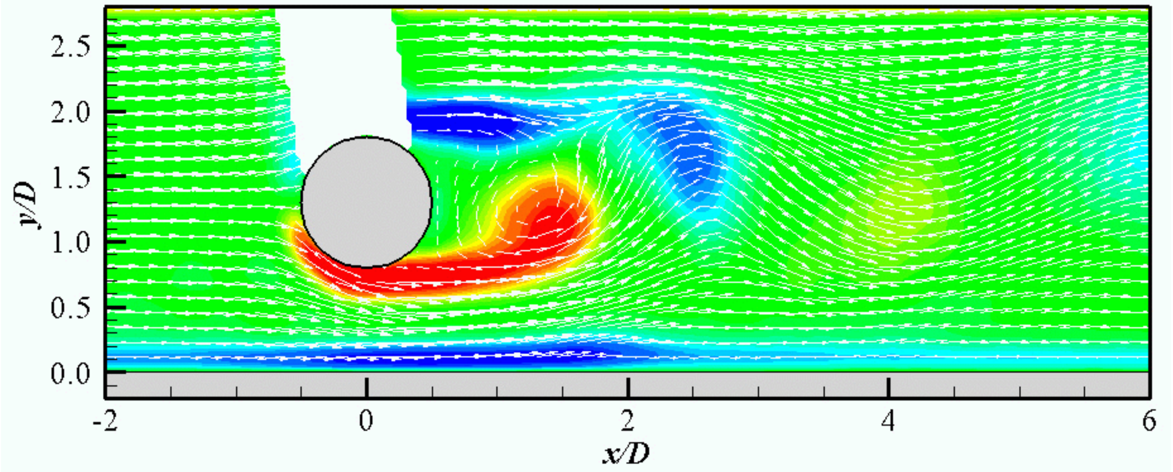
$$\boldsymbol{f}: \mathcal{D} \to \mathbb{R}^n$$

The vector field of gravitational field



The vector field of rotating object



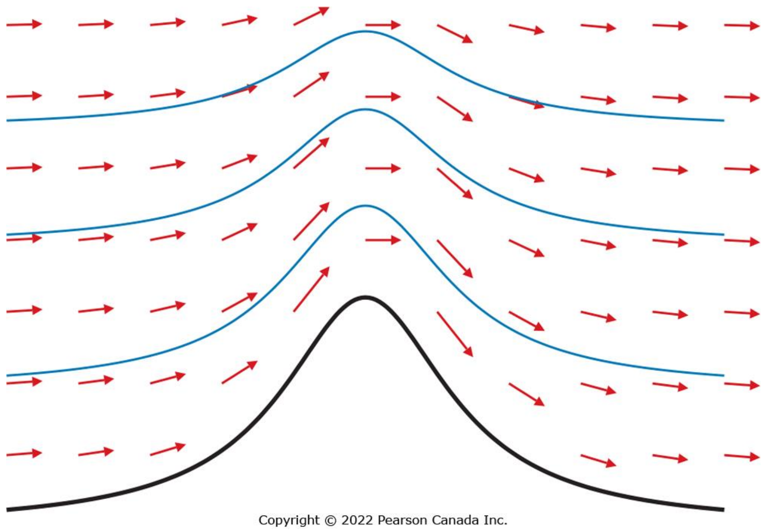


Field lines

Filed lines

For the vector field $\boldsymbol{f}: \mathcal{D} \to \mathbb{R}^d$ its field-lines/ steam-lines are family of curves $\boldsymbol{r}(t) = \sum_{i=1}^{d} x_i\ \hat{\boldsymbol{e}}_{x_i}$described by parameter t, such that on any point of the curve, the field $\boldsymbol{f}$ is tangent to the curve. i.e.,

$$\frac{d\boldsymbol{r}}{dt}(t) = \lambda(t)\boldsymbol{f}(\boldsymbol{r}(t))$$



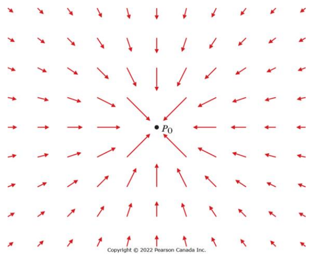
This yields the equation for all $1 \le i \le d$

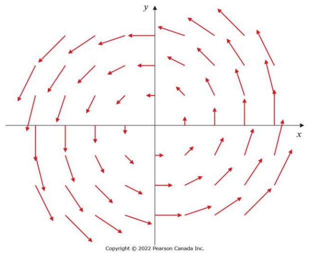
$$\frac{dx_i}{f_i(\boldsymbol{r})} = \lambda(t)dt$$

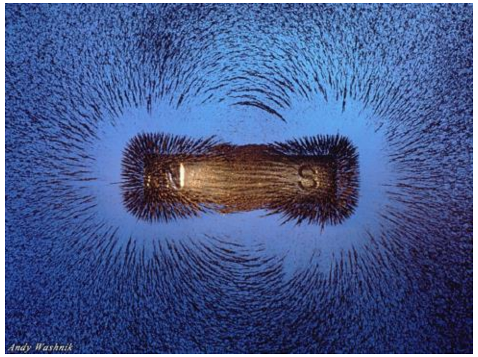
Question 2

What are the field lines of

1. The electric field of a point-charge?
2. A rotating object with angular velocity ω (a rotational field)?





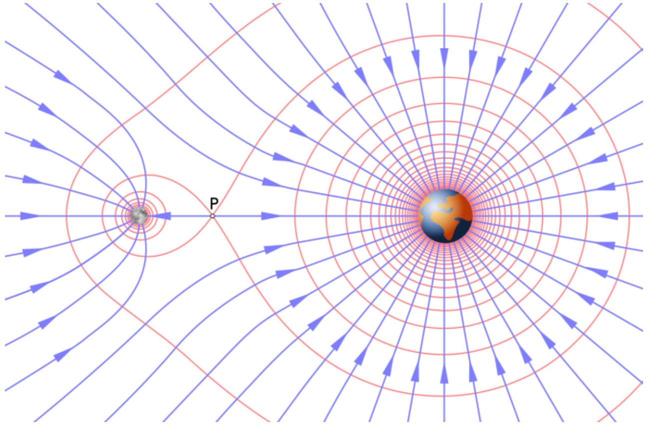


Conservative fields

Conservative field

If $\boldsymbol{f} = \nabla\varphi$, for some function $\varphi\colon \mathcal{D} \to \mathbb{R}$, then we call $\boldsymbol{f}$ a conservative field and φ a scalar potential.

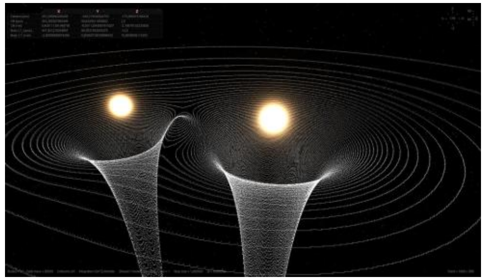
- They are conservative because the energy integral $\oint \boldsymbol{f} \cdot d\boldsymbol{x} = 0$ over any closed loop in $\mathcal{D}$ (to be understood in a later context)



Examples

- Gravitational, electric, magnetic fields are all conservative.
- <u>*The inverse square law*</u>: For point masses/charges q at $\boldsymbol{x} = \boldsymbol{x_0}$, the potential has the form $\varphi = Cq/|\boldsymbol{x} - \boldsymbol{x_0}|$ for some constant $C > 0$, and the field is $\boldsymbol{f} = -Cq(\boldsymbol{x} - \boldsymbol{x_0})/|\boldsymbol{x} - \boldsymbol{x_0}|^3$ with $|\boldsymbol{f}| = Cq/|\boldsymbol{x} - \boldsymbol{x_0}|^2$
- For n different masses/charges q_i at $\boldsymbol{x}_i$, the potential is

$$\varphi = \sum_{i=1}^{n} \frac{Cq_i}{|\boldsymbol{x} - \boldsymbol{x}_i|}$$



Run **Potential.py** to find the potential due to two point charges

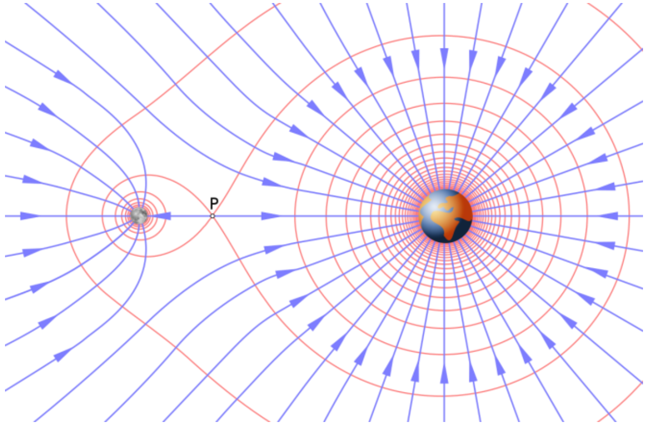
Equipotential surfaces

Conservative field

If $\boldsymbol{f} = \nabla\varphi$, for some function $\varphi: \mathcal{D} \to \mathbb{R}$, then we call $\boldsymbol{f}$ a conservative field and φ a scalar potential.

Equipotential surfaces

For a conservative field $\boldsymbol{f} = \nabla\varphi$, with potential $\varphi: \mathcal{D} \to \mathbb{R}$, the level-sets of φ are called equipotential surfaces.



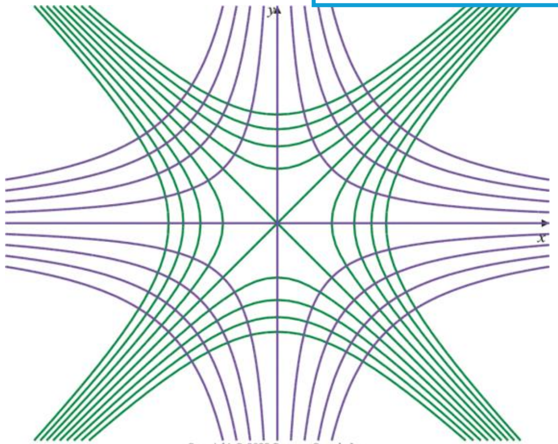
Question 3

Find the potential and equipotential surfaces of the field

$$\boldsymbol{f}(x,y) = \frac{\left(\frac{x}{a^2}\right)\hat{\imath} + \left(\frac{y}{b^2}\right)\hat{\jmath}}{\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^{3/2}}$$

$$\boldsymbol{f} = x\,\hat{\boldsymbol{\imath}} - y\,\hat{\boldsymbol{\jmath}}$$

$$\varphi = \frac{1}{2}(x^2 - y^2) + c$$



Conservative fields

Conservative field

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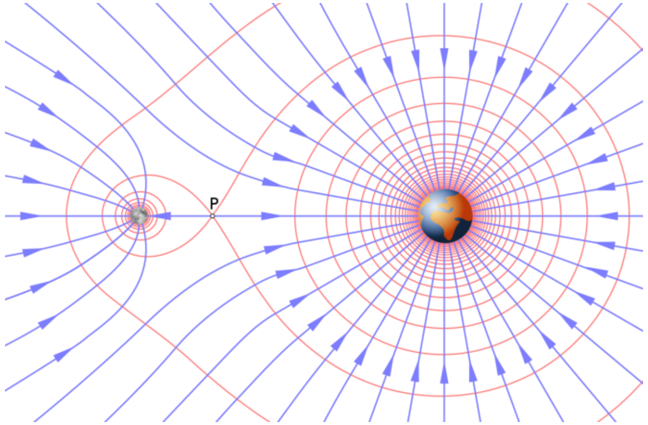

Theorem

Let $d \in \{2,3\}$, and $\mathcal{D} \subset \mathbb{R}^d$ be simply connected. Then a differentiable field $\boldsymbol{f}: \mathcal{D} \to \mathbb{R}^d$ is conservative if and only if for all $1 \leq i, j \leq d$,

$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_j}{\partial x_i};$$

in other words, if $D\boldsymbol{f}$ is symmetric.

Question 4.1

Show that the rotational field is not a conservative field

Question 4.2

Verify that the following fields are conservative and find their potential:

1. $\boldsymbol{f} = f_1(x)\hat{\boldsymbol{\imath}} + f_2(y)\,\hat{\boldsymbol{\jmath}} + f_3\,(z)\,\hat{\boldsymbol{k}}$
2. $\boldsymbol{f} = (2xy - z^2)\hat{\boldsymbol{\imath}} + (2yz + x^2)\hat{\boldsymbol{\jmath}} - (2zx - y^2)\,\hat{\boldsymbol{k}}$